CHE 2201 - Physical Chemistry II.

- Reaction kinetic
- Phase equilibrium.
- Quantom chemistry.

Reaction kinetic

$A + B \longrightarrow C$ at T, P normally happens under atmospheric pressure & ambient temperature. But sometimes can vary

01) Will reaction take place under a given state conditions.

$\Delta G = \Delta H - T\Delta S$

ΔH: enthalpy change - ; ΔS: entrophy change

$\Delta G < 0$ no reaction

$\Delta G > 0$ happen the reation

$\Delta G = 0$ in the equilibrium

02) How fast the reaction takes place

$N_2 + 3H_2 \rightleftharpoons 2NH_3$ yield < low / high

heating of water; hardoing of concrete, boiling of an egg } happen in a very short time

corosion (oxidation of iron), fading of colours of cloths } take long time

[At which rate or speed the reation takes place]

03) The extent of the reaction (yield of the product)

$Ag_{(aq)} + Cl^-_{(aq)} \longrightarrow AgCl_{(s)}$ → can get 100% results.

1 mol, 1 mol, 1 mol

$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ → can't get 100% of results.

1 mol, 3 mol, 2 mol (the yield is poor)

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equilibrium constant $\rightarrow K = \frac{P^2_{NH_3}}{P_{N_2} \times P^3_{H_2}}$

if the $k \rightarrow 10^{-10} - 10^{-15}$ - very slow

$10^{10} - 10^{20}$ - very speed

04) path by which the overall reaction occurs.

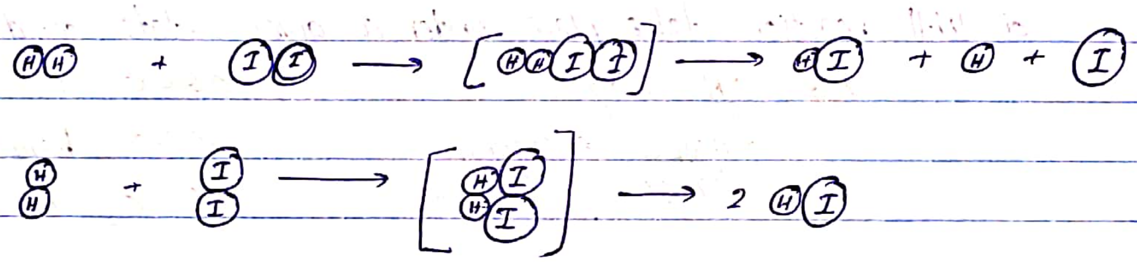
path means reaction mechanism.

* there are several requirements to take place reations.

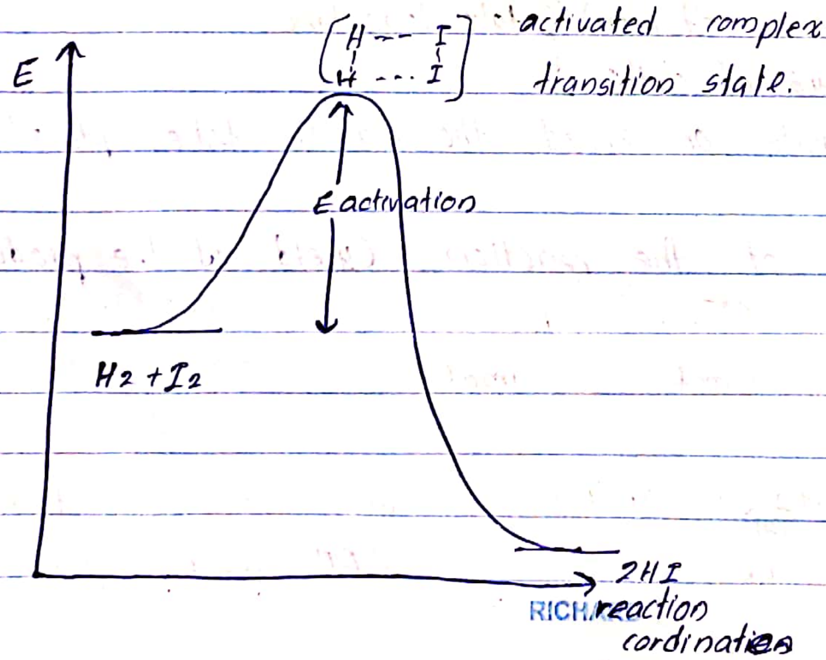
$H_{2(g)} + I_{2(g)} \longrightarrow 2HI_{(g)}$

according to kinetic molecular theory;

In the sample of gas, the gas molecule collid with each other.



Collision is not enough to happen a reaction. Also require reactant molecules collied in a favouarable orientation. (will only give the product).



* Activation energy; energy profile, energy required for the reactents to form the activated complex or transition state.

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Types of reactions

(01) Elementary reactions - Reactions takes place in a single step.

ex; $A + B \longrightarrow C$

Molecularity :- the number of reactant molecules in an elementry reaction.

$H_{2(g)} + I_{2(g)} \longrightarrow 2HI_{(g)}$

molecularity of $H_2 = 1$

$I_2 = 1$

$HI = 2$

$O_{3(g)} \longrightarrow O_{2(g)} + O_{(g)}$

also an elemantry reaction. but experimentally it has been found the reaction occurs in two steps.

But most of the reactions will place in many steps.

$O_{3(g)} \longrightarrow O_{2(g)} + O_{(g)}$

1st 2 $O_{3(g)}$ molecules called; $O_{3(g)} + O_{3(g)} \longrightarrow \underset{\text{activated molecule.}}{O_3^{*}} + O_3$

then activated O_3^{*} divided in to O_2, $O_{3(g)}^{*} \longrightarrow O_{2(g)} + O_{(g)}$

$2N_2O_{5(g)} \longrightarrow 4NO_2 + O_2$

$N_2O_5 \rightarrow NO_2 + NO_3$

$NO_3 \longrightarrow NO_2 + O_2$

$+ \quad NO + N_2O_5 \longrightarrow 3NO_2$

$2N_2O_5 \longrightarrow 4NO_2 + O_2.$

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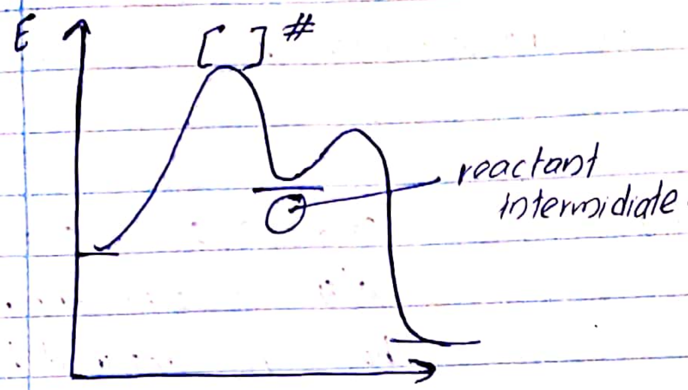
Reaction mechanism; detailed description of the sequence of individual steps that bring about the overall reaction.

Reaction intermidiate; this is not the activated complex.



- activated complex is unstable, because it has paritally broken bonds & made bonds.
- But the reactant intermidiate is stable.

- Activated complex can't isolated from the sample (reaction mix) because it's life time is very small.
- But reactant intermidiate can isolated from the reaction mixture.

kinetic → study of the rate of reaction and the mechanism a chemical reaction.

$A + B \rightarrow$ products.

A & B are consumed & products are formed.

Rate of the reaction → How fast the reactants are dissappeared or the products are formed.

$A + B \rightarrow D$ — this takes place in a constant volume.

$$Rate = -\frac{dn_A}{dt} = -\frac{dn_B}{dt} = +\frac{dn_D}{dt}$$

$$= -\frac{dn_A/v}{dt} = -\frac{dn_B/v}{dt} = \frac{dn_D/v}{dt}$$

$$= -\frac{dC_A}{dt} = -\frac{dC_B}{dt} = \frac{dC_D}{dt}$$

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$t_0 \rightarrow [A]_0$ average rate $\rightarrow \dfrac{\Delta C_A}{\Delta t} = \dfrac{[A]_2 - [A]_1}{t_2 - t_1}$ negative (-)

$\leftarrow$ Always $[A]_2 < [A]_1$

$t_1 \rightarrow [A]_{t_1}$

$\leftarrow$ Always $t_2 > t_1$

positive (+)

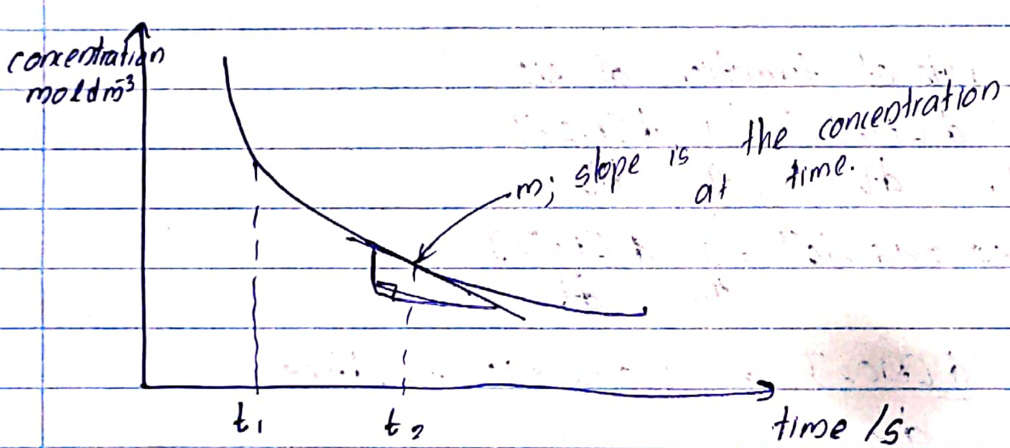
$t_2 \rightarrow [A]_{t_2}$

$= -\dfrac{\Delta C_A}{\Delta t}$

for B $\rightarrow -\dfrac{\Delta C_B}{\Delta t}$

for D $\rightarrow +\dfrac{\Delta C_D}{\Delta t}$

The rate at the instance at interest is the needed in the chemistry (instant average rate).



Factors that affect the rate of a reaction;

01) Concentrations of reactants
02) Temperature
03) Presence of catalyst
04) Presence of electromagnetic radiation (error) [light]

$H_2O_2 \rightarrow O_2 + H_2O$.

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$2N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$

The concentration of N_2O_5 after

$600\,s \rightarrow 1.25 \times 10^{-2}\ mol\,dm^{-3}$

$1200\,s \rightarrow 0.94 \times 10^{-2}\ mol\,dm^{-3}$

Find the rate of decomposition of N_2O_5;

$$Rate = \frac{C_t - C_o}{t_t - t_o} = \frac{0.94 \times 10^{-2} - 1.25 \times 10^{-2}}{1200 - 600}$$

$$= -\frac{0.31 \times 10^{-3}}{600}$$

$$= -0.0516 \times 10^{-4}\ mol\,dm^{-3}s^{-1}$$

rate of decomposition $= 0.0516 \times 10^{-4}\ mol\,dm^{-3}s^{-1}$

Find the rate of formation of NO_2.

$$\frac{1}{2} \frac{d[N_2O_5]}{dt} = \frac{1}{4} \frac{d[NO_2]}{dt}$$

$$\frac{1}{2} \times 0.0516 \times 10^{-4} = \frac{1}{4} \frac{d[NO_2]}{dt}$$

$$\frac{d[NO_2]}{dt} = 1.032 \times 10^{-5}\ mol\,dm^{-3}s^{-1}$$

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Concentration

When the concentration of reactants increases, rate of reaction increases.

When the concentration increases, number of molecules increases. There fore number of favourable collisions increases.

* All the collisions donot provide reaction.

$$A + B \longrightarrow \text{products}$$

$$-\frac{dC_A}{dt} = -\frac{dC_B}{dt} = \text{rate}$$

Rate of a reaction at a given temperature is directly proposional to the product of concentration term of each reactant, raise to a small power:

$$\text{rate} \propto C_A^{x} C_B^{y}$$

$$\boxed{\text{rate} = k\ C_A^{x}\ C_B^{y}}$$ ← Differential rate law.

C_A; Concentration of A
C_B; Concentration of B
x; Order of the reaction with respect to A
y; Order of the reaction with respect to B.
$x+y$; Overall order of the reaction.
k; Rate constant

* x and y are not equal to the steochemitric coefficients of the reactants in the balanced chemical equation. ∴ x & y are determine experimently. But in elementry reaction (single step) we can take those coefficients as x and y.

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0^{th} order

$A \rightarrow B$

this is zero order. so $x+y=0$. This reaction has only 1 reactant. $\therefore x=0$

$$-\frac{d[A]}{dt} = k[A]^0$$

$$-d[A] = k\,dt$$

when $t=0$ $\quad [A] = C_0$

$t=t$ $\quad [A] = C_t$

$$-\int_{C_0}^{C_t} d[A] = k\int_0^t dt$$

$$-[A]_{C_0}^{C_t} = k(t)_0^t$$

$$-(C_t - C_0) = k(t-0)$$

$$-C_t + C_0 = kt \quad \text{---} ①$$

$$C_t = -kt + C_0$$

$$y = -mx + c$$

Units of k; $k = -\frac{dC_A}{dt}$

$= \frac{mol\,dm^{-3}}{s}$

$= mol\,dm^{-3}\,s^{-1}$

Half life,

$t_{1/2}$; $C_0/2$

Then $C_t = C_0/2$

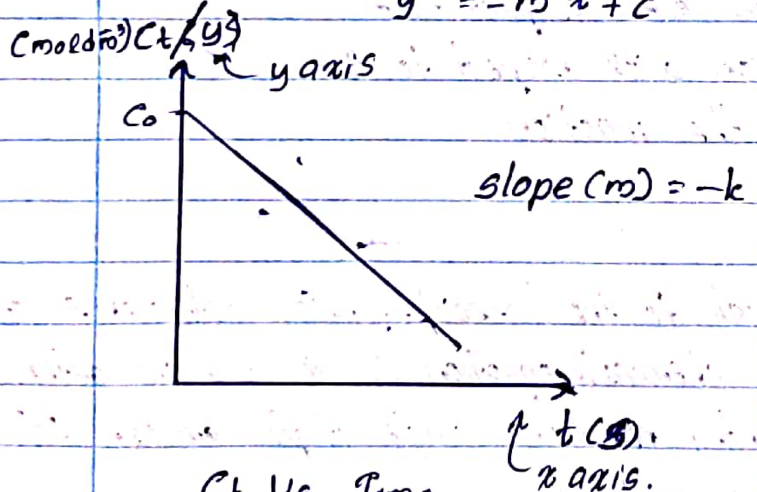
$t = t_{1/2}$

From ①;

$$-\frac{C_0}{2} + C_0 = k\,t_{1/2}$$

$$t_{1/2} = \frac{C_0}{2k}$$

✱ In 0^{th} order reaction half life depends on the concentration.



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$\ln(x) + \ln(y) = \ln(xy)$

$\ln(x) - \ln(y) = \ln(x/y)$

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1^{st} order

$A \rightarrow B$

In first order $x = 1$

$$-\frac{d[A]}{dt} = k[A]$$

$t = 0 ; [A] = C_0$

$t = t ; [A] = C_t$

$$-\int_{C_0}^{C_t} \frac{1}{[A]} d[A] = k \int_0^t dt$$

$$-\left(\ln[A]\right)_{C_0}^{C_t} = k(t-0)$$

$$\cancel{-[\log(C_t) - \log(C_0)] = kt}$$

$$\cancel{-\log(C_t) + \log(C_0) = kt}$$

$$\cancel{\log C_t = -kt}$$

$$-[\ln(C_t) - \ln(C_0)] = kt$$

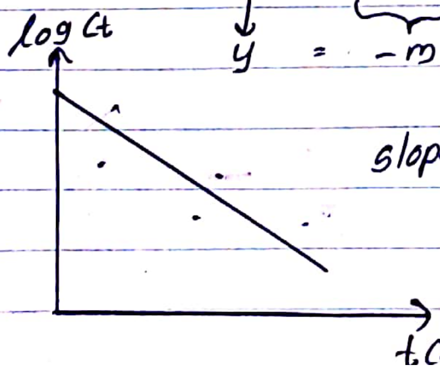
$$-\ln(C_t) + \ln(C_0) = kt$$

$$-2.303\log(C_t) + 2.303\log(C_0) = kt$$

$$-\log C_t + \log C_0 = \frac{kt}{2.303}$$

$$\underset{y}{\log C_t} = \underbrace{-\frac{k}{2.303}}_{-m} \underset{x}{t} + \underset{c}{\log C_0}$$

$y = -m x + c$



$$\text{slope}(m) = \frac{-k}{2.303}$$

$\ln(x) = 2.303 \log(x)$ / $\int \frac{1}{x} dx = \ln(x)$

Units of k;

$$k = \frac{-d[A]}{dt} \times \frac{1}{[A]}$$

$$= \frac{mol\,dm^{-3}}{s} \times \frac{1}{mol\,dm^{-3}}$$

$$= s^{-1}$$

Half life;

$t_{1/2}$; $C_0/2$

$C_t = C_0/2$

$$\ln C_t - \ln C_0 = -k t_{1/2}$$

$$\ln\left(\frac{C_t}{C_0}\right) = -k t_{1/2}$$

$$\ln\left(\frac{C_0/2}{C_0}\right) = -k t_{1/2}$$

$$\ln(1/2) = -k t_{1/2}$$

$$t_{1/2} = \frac{-\ln(1/2)}{k}$$

$$\ln(1/2) = 0.693$$

$$t_{1/2} = \frac{0.693}{k}$$

$t_{1/2}$ dosen't depend on the concentration.

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Second order

$$A \rightarrow B.$$

In second order $x = 2$.

$$-\frac{d[A]}{dt} = k[A]^2$$

$t = 0;\ [A] = C_0$

$t = t;\ [A] = C_t$

$$-\int_{C_0}^{C_t} \frac{1}{[A]^2}\, d[A] = k\int_0^t dt$$

$$-\left(\frac{1}{[A]}\right)_{C_0}^{C_t} = k(t-0)$$

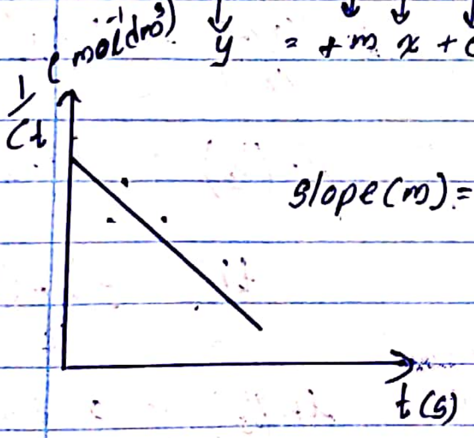
$$+\left(\frac{1}{C_t} + \frac{1}{C_0}\right) = kt$$

$$+\frac{1}{C_t} \mp \frac{1}{C_0} = kt$$

$$\frac{1}{C_t} = +kt + \frac{1}{C_0}$$

$$\downarrow \quad \downarrow\ \downarrow \quad \downarrow$$

$$y = +m\,x + C$$



have a positive gradient.

Units of k;

$$k = -\frac{d[A]}{dt} \times \frac{1}{[A]^2}$$

$$= \frac{mol\,dm^{-3}}{s} \times \frac{1}{mol^2\,dm^{-6}}$$

$$= mol^{-1}\,dm^3\,s^{-1}$$

Half life;

$$\frac{1}{C_t} = kt + \frac{1}{C_0}$$

$$\frac{2}{C_0} = kt_{1/2} + \frac{1}{C_0}$$

$$kt_{1/2} = -\frac{1}{C_0} + \frac{2}{C_0}$$

$$kt_{1/2} = \frac{1}{C_0}$$

$$t_{1/2} = \frac{1}{C_0 k}$$

half life is depend on concentration.

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Methods of determine the order of the reaction.

Initial rate method
Isolation method
Half life time method

Initial rate method

$$A + B \longrightarrow \text{products}$$

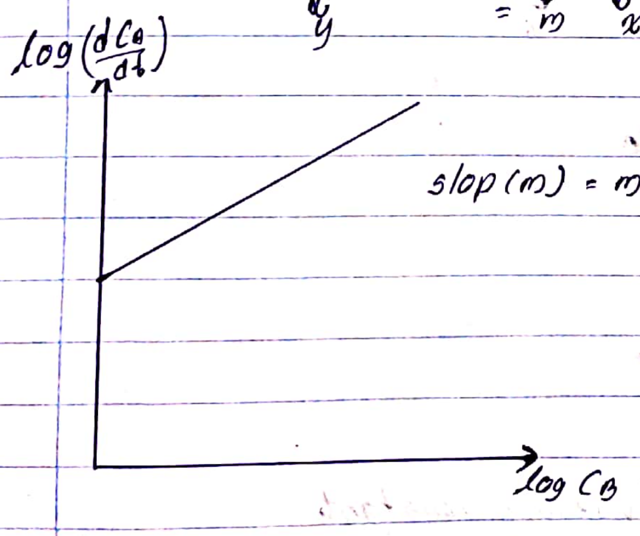
$$-\frac{dC_A}{dt} = -\frac{dC_B}{dt} = kC_A^{n}C_B^{m}$$

Here we keep the concentration of one reactant in constant level and change the concentrations of other reactant and measure the reactant rate.

$$\log -\left(\frac{dC_A}{dt}\right) = \log k + n\log C_A + m\log C_B$$

keep [A] constant;

$$\underbrace{\log -\left(\frac{dC_A}{dt}\right)}_{y} = \underbrace{m}_{m}\,\underbrace{\log C_B}_{x} + \underbrace{\log k + n\log C_A}_{c}$$



like this we can keep [B] constant and draw the graph.

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Isolation method.

Keep concentration of one reactant in large excess. So in the time duration of reaction, that reactant change in very (too) small and don't consider. So other reactant is change and can see.

$$A + B \longrightarrow \text{product}$$

$$\text{rate} = k\, C_A^{\alpha} C_B^{\beta}$$

Case (01); if $C_B \ggg C_A$;

changing C_B is negligible. But C_A is decreasing. C_B remains virtually constant.

$$\text{rate} = k\, C_A^{\alpha} C_B^{\beta}$$

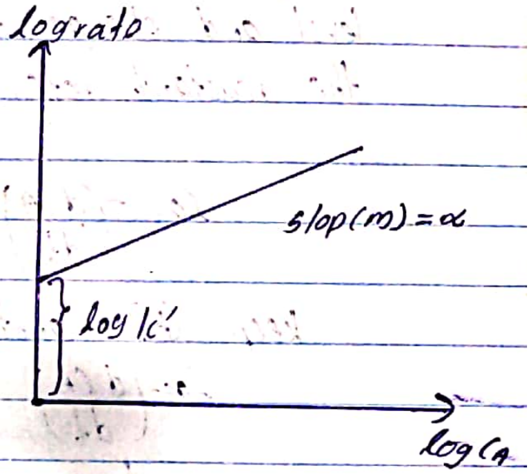
$$k' = k\, C_B^{\beta}$$

$$\text{rate} = k' C_A^{\alpha}$$

$$\log \text{rate} = \log k' + \log C_A^{\alpha}$$

$$\log -\frac{dA}{dt} = \log k' + \alpha \log C_A$$

$$y = c + m\, x$$



Case (02); if the $C_A \ggg C_B$

$$\text{rate} = k\, C_A^{\alpha} C_B^{\beta}$$

$$k'' = k\, C_A^{\alpha}$$

$$\text{rate} = k'' C_B^{\beta}$$

Concentration of A remains virtually constant.

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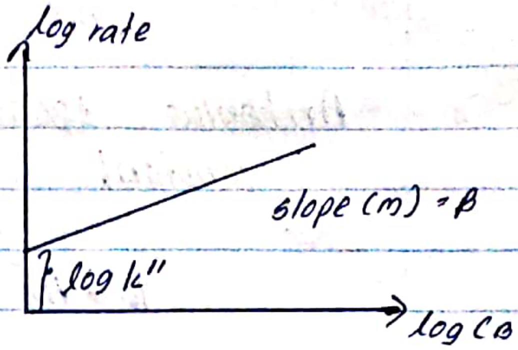
$$\log \text{rate} = \log k'' + \log C_B^{\beta}$$

$$\log \text{rate} = \log k'' + \beta \log C_B$$

$$y = c + m x$$



pseudo ~~first o~~ alpha order; is a term used to describe a reaction that appears to follow first order kinetics under certain conditions, even though the reaction actually involves more than one reactant. The reaction rate depends primarily on the concentration of other reactants (we make ones concentration constant), and the overall rate law can be treated as if the reaction were first-order with respect to other limiting reactant.

pseudo beta order reaction is a term used to describe a reaction that appears to follow second order kinetics under certain conditions, even though the reaction may involve more than one reactant. This situation typically occurs when one of the reactants is in much higher concentration than the others.

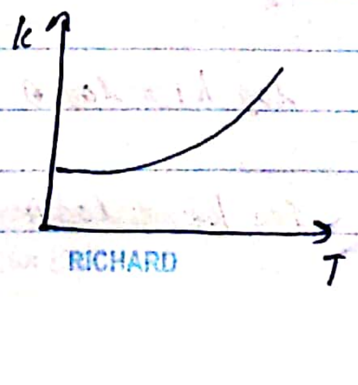
If there are three reactants in the reaction,

$$A + B + C \longrightarrow \text{products.}$$

should stay B and C as a large amount and isolate A.

Temperature

Temperature has significant effect on the rates of chemical reactions. As the temperature increases the rate increase exponentially.



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Arrhenius Equation / Quantitative estimation of rate constant.

$$k = Ae^{-Ea/RT}$$

- A - pre exponential factor (frequency factor)
- Ea - activation energy
- R - gas constant
- T - absolute temperature
- k - rate constant.

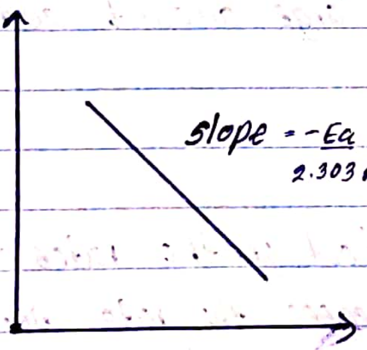
$\frac{Ea}{RT}$ → fraction of molecules called in proper orientation with the activation energy increases.

Activation energy related to the reaction constant.

$$\ln k = \ln A - \frac{Ea}{RT}$$

$$\underset{\downarrow}{\log k} = \underset{\downarrow}{\log A} - \underset{\downarrow}{\frac{Ea}{2.303\,R}}\ \underset{\downarrow}{\frac{1}{T}}$$

$$y = c - m\ x$$



can determine the activation energy of a reaction.

at $T_1 \rightarrow k_1$; $\log k_1 = \log A - \frac{Ea}{2.303\,R}\ \frac{1}{T_1}$ —— ①

$T_2 \rightarrow k_2$; $\log k_2 = \log A - \frac{Ea}{2.303\,R}\ \frac{1}{T_2}$ —— ②

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$$① - ② \quad \log k_1 - \log k_2 = \log A - \frac{Ea}{2.303R}\frac{1}{T_1} - \left(\log A - \frac{Ea}{2.303R}\frac{1}{T_2}\right)$$

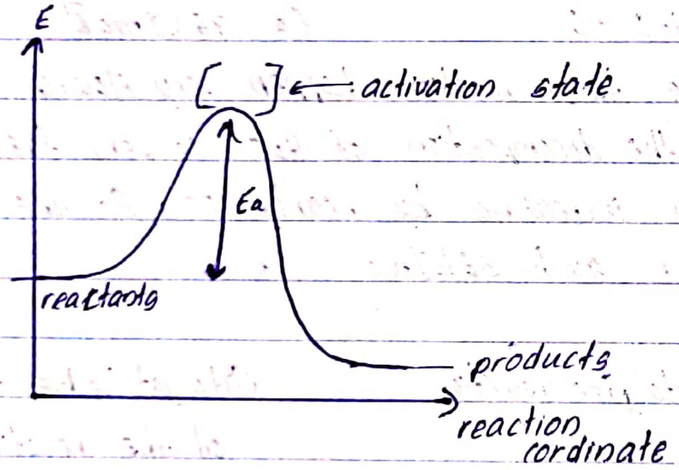
$$\log\left(\frac{k_1}{k_2}\right) = \log A - \frac{Ea}{2.303R}\frac{1}{T_1} - \log A + \frac{Ea}{2.303R}\frac{1}{T_2}$$

$$\log\left(\frac{k_1}{k_2}\right) = \frac{Ea}{2.303R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$Rate = k\ C_A^{\alpha} C_B^{\beta}$$

$$k = A e^{-Ea/RT}$$

$$Rate = A e^{-Ea/RT} C_A^{\alpha} C_B^{\beta}$$



To happen a reaction the reactants have to reach activation energy.

Catalysts.

When the activation energy is very high, the rate of is very law at moderate temperature.

moderate temperature; little higher than the room temperature.

Increasing the temperature is not always increase the rate. specially industrial scale reactions.

high Temperature → high operational cost.

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Catalyst increase the rate of a reaction at moderate temperature.

Catalyst

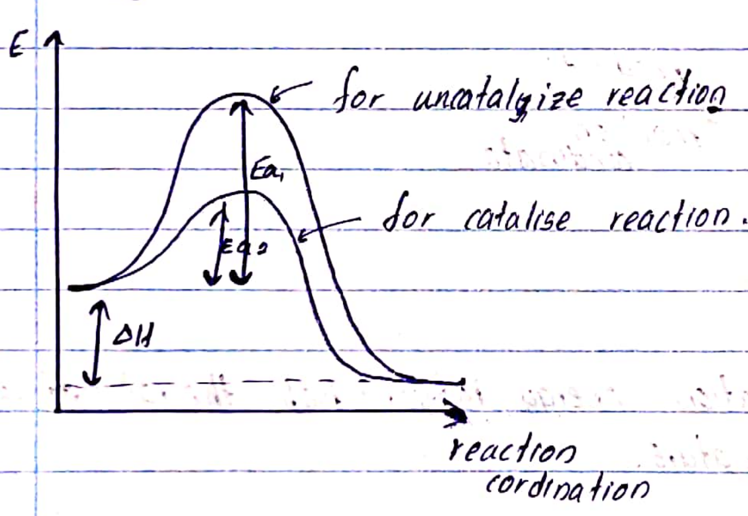
Homogeneous
All the reactants, products and catalyst are in same phase

Heterogeneous
The catalyst in a different phase when compared with reactants and products.

How a catalyst increase the rate of a reaction.

$$2H_2O_2 \longrightarrow 2H_2O + O_2 \qquad Ea = 76\,kJmol^{-1}$$

At room temperature, this reaction happen very slowly.
But presense in KI, the decomposition of H_2O_2 is very fast.
Rate of the reaction increasing by nearly 200. So the activation energy is become lower such $56\,kJmol^{-1}$



Catalyst reduse the Eacti of the reaction.
But dosen't change the enthalphy change of the reaction.

$$A + B \rightleftharpoons C$$

equilibrium constant. → $k = \dfrac{[C]}{[A]\,[B]}$

Catalyst do not affect on the equilibrium constant. Catalyst change the rate of forward reaction and the backword reaction.

* To a reaction it is needed only few amount of catalysts.

How the catalyst change the rate of reaction.

ex 01:-
$$H_2O_2 + I^- \xrightarrow{slow} H_2O + OI^-$$ ← determine the overall rate of the reaction
$$H_2O + OI^- \xrightarrow{fast} H_2O_2 + O_2 + I^-$$

$$rate = k[H_2O_2][I^-]$$

ex 02:- Bromination of acetone.

this is a very slow reaction, Because Ea is very high at room temperature. But in a presense of a base, the reaction became fast.

$$CH_3COCH_3 + OH^- \xrightarrow{slow} CH_3COCH_2^- + H_2O$$

base can abstract the H^+ ion from acetone.

$$CH_3COCH_2^- + Br_2 \xrightarrow{fast} CH_3COCH_2Br + Br^-$$

$$Rate = k\ [CH_3COCH_3]^x [OH^-]$$

The rate dosen't depend on Br because it participade in fast rxn.

Catalysts provide a new path to the reaction which avoids the rate determining step. It changes the mechanism. to a path that has low activation energy.

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Electro magnetic radiation.

Electro magnetic radiation have both electro and magnetic field. This can interact with matter.

$$m \xrightarrow{h\nu} m^*$$

light; molecule absorb light and goes to exited states.
m^* is very reactive

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Reaction mechanisms

Detailed description of the sequence of individual step that lead to overall reaction.

Mechanism of a reaction is determing by finding the rate low or order of the reaction with respect to different reactants.

Mechanism can be tested by two ways.

1) mechanism should predict correctly the amout and nature of products,

2) by determining individual rate constant of the different test involved.

Features of the reaction.

rate determining step - slowest step of the reaction mechanism

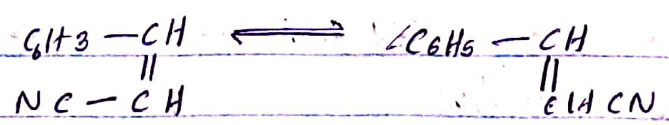
Reversible first order reaction.

$$A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B$$

reaction proceeds on both directions at equilibrium

rate of forward reaction = rate of backword reaction.

The net rate of formation of each pices is zero



cis - trans izomerization

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$$A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B$$

The rate change of A,

$$\frac{-d[A]}{dt} = -k_1[A] + k_{-1}[B] \text{ —— } ①$$

$t=0$, $[A] = [A]_0$ $[B] = 0$

$t=t$, $[A] = [A]$ $[B] = [B]$

$$[B] = [A]_0 - [A] \text{ —— } ②$$

from ① and ②,

$$\frac{-d[A]}{dt} = -k_1[A] + k_{-1}([A_0] - [A])$$

$$= k_{-1}[A_0] - (k_1 + k_{-1})[A] \text{ —— } ③$$

at equilibrium

$$\frac{d[A]}{dt} = 0 \qquad [A] = [A]_{eq}$$

$$k_{-1}[A_0] - (k_1 + k_{-1})[A]_{eq} = 0$$

$$k_{-1}[A_0] = (k_1 + k_{-1})[A]_{eq}$$

$$[A_0] = \left(\frac{k_1 + k_{-1}}{k_{-1}}\right)[A]_{eq} \text{ —— } ④$$

from ④ and ③,

$$\frac{-d[A]}{dt} = k_{-1}\left(\frac{k_1 + k_{-1}}{k_{-1}}\right)[A]_{eq} - (k_1 + k_{-1})[A]$$

$$\frac{-d[A]}{dt} = (k_1 + k_{-1})[A]_{eq} - (k_1 + k_{-1})[A]$$

$$\frac{-d[A]}{dt} = (k_1 + k_{-1})([A]_{eq} - [A])$$

$$\frac{-d[A]}{([A]_{eq} - [A])} = (k_1 + k_{-1})\,dt$$

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$$\cancel{t=0 \quad [A] = [A]_0}$$
$$\cancel{t=t \quad [A] = [A]}$$

From backword reaction & ④,

$$\frac{d[A]}{dt} = k_{-1}\left(\frac{k_1+k_{-1}}{k_{-1}}\right)[A]_{eq} - (k_1+k_{-1})[A]$$

$$\frac{d[A]}{dt} = -(k_1+k_{-1})([A]-[A]_{eq})$$

$$-\cancel{\frac{d[A]}{dt}} \quad \frac{-d[A]}{[A]-[A]_{eq}} = (k_1+k_{-1})dt$$

intergrade at $t=0$ $[A] = [A]_0$
$t=t$ $[A] = [A]$

$$\frac{\int d[A]}{[A]-[A]_{eq}} = (k_1+k_{-1})\int dt$$

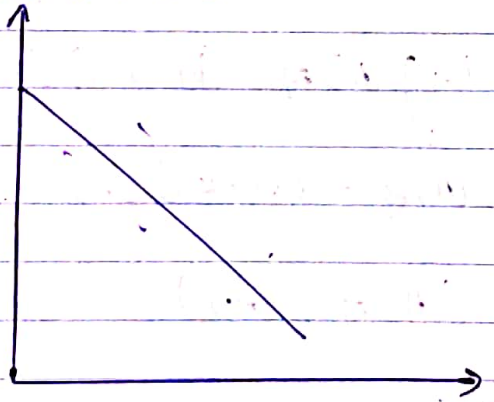
$$\ln\left(\frac{[A]-[A]_{eq}}{[A]_0-[A]_{eq}}\right) = (k_1+k_{-1})t \quad ⑤$$

$$\ln([A]-[A]_{aq}) - \ln([A]_0-[A]_{eq}) = (k_1+k_{-1})t$$

$$\ln([A]-[A]_{eq}) = (k_1+k_{-1})t + \ln([A]_0-[A]_{aq})$$

$$y = m \quad x + c$$

$\ln([A]-[A]_{aq})$



$t(s)$

slope $= k_1 + k_{-1}$

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$$\frac{k_1}{k_{-1}} = \frac{[B]_{eq}}{[A]_{eq}} = \frac{[A]_0 - [A]_{eq}}{[A]_{eq}}$$

$$[A]_{eq} = \frac{k_1}{k_1 + k_{-1}} [A]_0$$

$$\frac{\Delta[A]}{\Delta[A]_0} = ln - (k_1 + k_{-1})t$$

$$\Delta[A] = \Delta[A]_0 \, e^{-(k_1+k_{-1})t}$$

Consecutive first order reaction.

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

$$-\frac{d[A]}{dt} = -k_1[A] \quad \text{——— ①}$$

$$\frac{d[B]}{dt} = k_1[A] - k_2[B] \quad \text{——— ②}$$

$$\frac{d[C]}{dt} = k_2[B] \quad \text{——— ③}$$

$t = 0$ $\quad [A] = [A]_0 \quad [B] = 0 \quad [C] = 0$

$t = t$ $\quad [A] = [A]_t \quad [B] = [B] \quad [C] = [C]$

from ①

$$\int \frac{d[A]}{[A]} = -k_1 \int dt$$

$$ln[A] = -k_1 t$$

$$ln[A]_0 - ln[A_t] = -k_1 t$$

$$\cancel{ln[A]_0 = -k_1 t + ln[A]_t}$$

$$\cancel{[A] = [A]_0 \, e^{k_1 t}} \quad \text{——— ④}$$

$$ln\frac{[A_0]}{[A_t]} = -k_1 t$$

$$(A_t) = (A_0) \, e^{-k_1 t} \quad \text{——— ④}$$

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To get concentration of B at time t,

$$\frac{d[B]}{dt} = k_1 [A_0] e^{-k_1 t} - k_2 [B]$$

rearranging intergrating,

$$[B] = \frac{k_1 [A]_0}{k_2 - k_1} (e^{-kt} - e^{-k_2 t}) \quad \text{——— (5)}$$

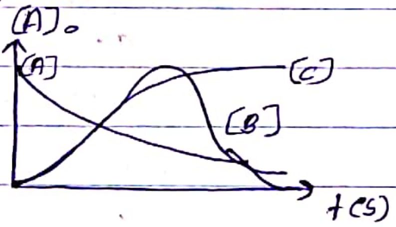
To find the concentration of C

$$[A]_0 = [A] + [B] + [C]$$

$$[C] = [A]_0 - [A] - [B]$$

$$[C] = [A_0] \left(1 - \frac{k_2 e^{-k_1 t}}{k_2 - k_1} + \frac{k_1 e^{-k_1 t}}{k_2 - k_1} \right)$$

$$[A] \xrightarrow[slow]{k_1} [B] \xrightarrow[fast]{k_2} [C]$$

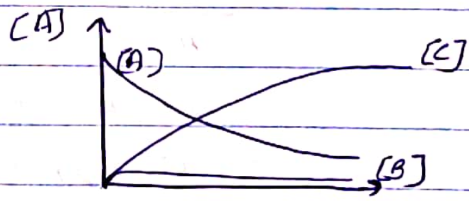


Graph shows concentration intermidiate B rises to a maximum and drop back to zero, as reaction as [A] depleted.

suppose $k_2 > k_1$

Then whenever B is form it's decases quickly to C the rate of formation of the product C is then determine by the rate at which B is formed.

$$e^{-k_1 t} \ggg e^{-k_2 t}$$



$$[C] = [A]_0 (1 - e^{-kt})$$

$$[B] = \frac{k_1}{k_2} [A]_0 e^{-kt}$$

The steady state$^{k_1 t}$ approximation

$$[B] = \frac{k_1}{k_2} [A]$$

When A is reacting slowly with in a short period of time A can bes assumed to be a constant. In the short period of time. and k_1/k_2 is very small. The concentration of B would be very small at any given time when compare to a

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The steady state approximation

In multiple reaction mechanisms one or more intermidiate species that do not appear in the overall equation are very reactive. Therefore do not accumilate to any significant extend during the reaction. So we can assume that the concentration of B rises to maximum and then fall back to zero.

$$H^+ + HNO_2 + C_6H_5NH_2 \xrightarrow{Br^-} C_6H_5N_2^+ + 2H_2O$$

proposed reaction mechanism,

$$H^+ + HNO_2 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} H_2NO_2^+ \quad (fast)$$

$$H_2NO_2^+ + Br^- \xrightarrow{k_2} ONBr + H_2O \quad (slow)$$

$$ONBr + C_6H_5NH_2 \xrightarrow{k_3} C_6H_5N^+ + H_2O + Br^- \quad (fast)$$

Observed rate law for this mechanism.

$$rate = k[H^+][HNO_2][Br^-] \quad \text{—①}$$ — we have to prove this

$$-\frac{d[C_6H_5NH_2^+]}{dt} = k_3[ONBr][C_6H_5NH_2] \quad \text{—②}$$

In here intermidiates are, $H_2NO_2^+$, $ONBr$

Applying the steady state approximation,

$$\frac{d[ONBr]}{dt} = 0 \quad \text{—③}$$

$$\frac{d[ONBr]}{dt} = k_2[H_2NO_2^+][Br^-] - k_3[ONBr][C_6H_5NH_2]$$

From ③,

$$0 = k_2[H_2NO_2^+][Br^-] - k_3[ONBr][C_6H_5NH_2]$$

$$k_3[ONBr][C_6H_5NH_2] = k_2[H_2NO_2^+][Br^-]$$

$$[ONBr] = \frac{k_2}{k_3}\frac{[H_2NO_2^+][Br^-]}{[C_6H_5NH_2]} \quad \text{—④}$$

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$$\frac{d[H_2NO_2^+]}{dt} = 0 \quad \text{———} \quad (5)$$

$$\frac{d[H_2NO_2^+]}{dt} = k_1[H^+][HNO_2] - k_{-1}[H_2NO_2^+] - k_2[Br^-][H_2NO_2^+]$$

From (5),

$$0 = k_1[H^+][HNO_2] - k_{-1}[H_2NO_2^+] - k_2[Br^-][H_2NO_2^+]$$

$$k_{-1}[H_2NO_2^+] + k_2[Br^-][H_2NO_2^+] = k_1[H^+][HNO_2]$$

$$[H_2NO_2^+](k_{-1} + k_2[Br^-]) = k_1[H^+][HNO_2]$$

$$[H_2NO_2^+] = \frac{k_1[H^+][HNO_2]}{(k_{-1} + k_2[Br^-])} \quad \text{———} \quad (6)$$

The rate is determing according to the slowest reaction,

$$Rate = k_2[Br^-][H_2NO_2^+] \quad \text{———} \quad (7)$$

From (6)

$$Rate = k_2[Br^-] \times \frac{k_1[H^+][HNO_2]}{k_{-1} + k_2[Br^-]}$$

Assume that, $k_{-1} \gg\gg\gg k_2$

$$Rate = \frac{k_2 k_1}{k_{-1}}[Br^-][H^+][HNO_2]$$

$$\frac{k_2 k_1}{k_{-1}} = k$$

$$Rate = k[Br^-][H^+][HNO_2]$$

Requirements,

- write a differential rate law to the rate determining step.
- eliminate the concentration of any reaction intermidiate.

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Pre equilibrium

$$A \underset{k_{-1}/k_2}{\overset{k_1}{\rightleftharpoons}} M + C$$

$$M + B \xrightarrow{k_3} P$$

The reaction intermoidiate is form by a reversible process thial to the final step of the reaction.

The pre equilibrium approximation assume that the reactants and intermoidiates of a multi step reaction are in dynamic equilibrium.

The equilibrium approaimatron requires the first step to be faster

$$\frac{-d[B]}{dt} = k_3 [B][M] \text{ --- } ①$$

Apply steady state approximation,

$$\frac{d[M]}{dt} = k_1[A] - k_2 [M][C] - k_3[M][B]$$

$$\frac{d[M]}{dt} = 0$$

$$0 = k_1[A] - k_2[M][C] - k_3[M][B]$$

$$k_2[M][C] + k_3[M][B] = k_1[A]$$

$$[M](k_2[C] + k_3[B]) = k_1[A]$$

$$[M] = \frac{k_1[A]}{k_2[C] + k_3[B]} \text{ --- } ②$$

From ① & ②;

$$\frac{-d[B]}{dt} = k_3[B] \times \frac{k_1[A]}{k_2[C] + k_3[B]}$$

$$= \frac{k_1k_3[B][A]}{k_2[C] + k_3[B]}$$

If, $k_2 >>> k_3$

$$\frac{-d[B]}{dt} = \frac{k_1k_3[B][A]}{k_2[C]} = \frac{k[B][A]}{[C]} \quad \left(k = \frac{k_1k_3}{k_2}\right.$$

If, $k_3 >>> k_2$

$$\cancel{k = \frac{k_1k_3}{k_2}} \qquad \frac{-d[B]}{dt} = k_1[A]$$

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$$2N_2O_5 \longrightarrow 4NO_2 + O_2$$

According to the theoritical rate law,

$$-\frac{d[N_2O_5]}{dt} = k[N_2O_5]^2$$

But according to the experimental rate law,

$$-\frac{d[N_2O_5]}{dt} = k'[N_2O_5]$$

This reaction goes through many steps,

$$N_2O_5 \xrightarrow{k_1} [NO_2][NO_3] \text{ --- } ①$$

$$NO_2 + NO_3 \xrightarrow{k_2} [N_2O_5] \text{ --- } ②$$

$$NO_2 + NO_3 \xrightarrow{k_3} NO_2 + O_2 + NO^- \text{ --- } ③$$

$$NO + N_2O_5 \xrightarrow{k_4} 3NO_2 \text{ --- } ④$$

$$\frac{d[NO_3]}{dt} = 0$$

$$\frac{d[NO_3]}{dt} = k_1[N_2O_5] - k_2[NO_2][NO_3] - k_3[NO_2][NO_3]$$

$$0 = k_1[N_2O_5] - k_2[NO_2][NO_3] - k_3[NO_2][NO_3]$$

$$k_1[N_2O_5] = [NO_3](k_2[NO_2] + k_3[NO_2]) \text{ --- } (01)$$

$$\frac{d[NO]}{dt} = k_3[NO_2][NO_3] - k_4[NO][N_2O_5] \text{ --- } ④$$

$$\frac{d[NO]}{dt} = 0$$

$$0 = k_3[NO_2][NO_3] - k_4[NO][N_2O_5]$$

$$k_3[NO_2][NO_3] = k_4[NO][N_2O_5]$$

$$[NO] = \frac{k_3[NO_2][NO_3]}{k_4[N_2O_5]} \text{ --- } (02)$$

According to (01) $[NO_3] = \frac{k_1[N_2O_5]}{k_2[NO_2] + k_3[NO_2]}$ --- (03)

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From ~~2~~ (02) and (03)

$$[NO] = \frac{k_3 [NO_2]}{k_4 [N_2O_5]} \times \frac{k_1 [N_2O_5]}{k_2[NO_2] + k_3[NO_2]}$$

$$= \frac{k_1 k_3}{k_4 (k_2 + k_3)} \quad \text{——— (5)}$$

The overall rate of the reaction is given by the rate of decomposition of N_2O_5.

$$\text{rate} = -\frac{d[N_2O_5]}{dt}$$

$$\cancel{-\frac{d[N_2O_5]}{dt} = -k_1 [N_2O_5] + k_2 [N_2O_5] - k_4 [N_2O_5][NO]}$$

$$-\frac{d[N_2O_5]}{dt} = -k_1[N_2O_5] + k_2[NO_2][NO_3] - k_4[N_2O_5][NO]$$

$$= -k_1[N_2O_5] + \frac{k_2 k_4 [NO][N_2O_5]}{k_3} - k_4[N_2O_5][NO]$$

$$= -k_1[N_2O_5] + [N_2O_5]\left(\frac{k_2k_4}{k_3} - k_4\right)\left(\frac{k_1k_3}{k_4(k_2+k_3)}\right)$$

$$= [N_2O_5]\left[-k_1 + \frac{(k_2\cancel{k_4} - \cancel{k_4}k_3)(k_1\cancel{k_3})}{\cancel{k_3}\cancel{k_4}(k_2+k_3)}\right]$$

$$= [N_2O_5]\left\{-k_1 + \frac{(k_2-k_3)\,k_1}{(k_2+k_3)}\right\}$$

$$= [N_2O_5]\left(\frac{\cancel{-k_1k_2} - k_1k_3 + \cancel{k_1k_2} - k_1k_3}{k_2+k_3}\right)$$

$$= [N_2O_5]\left(\frac{-2k_1k_3}{k_2+k_3}\right)$$

$$k' = \frac{-2k_1k_3}{k_2+k_3}$$

$$\therefore \text{rate} = k'[N_2O_5]$$

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Chain Reactions,

Features;

Radicals are formed. They are very reactive.

$Cl_2 \xrightarrow{h\nu} \dot{C}l$ ← radical.

They iniciate the reaction.

Explotions and polymerization reactions are chain reactions.

Some steps in chain reactions are cannot be control.

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